

SIMATS ENGINEERING
ASSESSMENT TOOL 3 – VIVA QUESTIONS

Course:	Code:	CO: CO5
Duration:	Total Marks: 100	Reg No / Name :

COURSE OUTCOME COVERED

CO5: *Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)*

Code of Conduct:

I _M Kavya 192472370_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kavya

Instructions: Answer all questions. Each question carries 4 marks.

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1. How would you evaluate the security posture of a Windows system after identifying multiple open services?

Answer:

List every open service and its version, check each against known vulnerability databases (CVE/Exploit-DB), determine which services are actually needed for the business, verify authentication requirements, and rate overall risk based on how exposed and exploitable the services are.

2. Which factors would you consider when choosing a hacking tool for Windows vulnerability assessment, and why?

Answer:

Consider the target OS/version compatibility, what exactly needs testing (scanning, exploitation, or enumeration), the tool's accuracy and false-positive rate, whether it is actively maintained/updated, and legal authorization to use it.

3. How would you evaluate the security risk associated with an exposed RPC service?

Answer:

Identify the RPC version running, check it against known RPC-related CVEs (e.g., MS-RPC exploits), determine if it's reachable from untrusted/external networks, and assess what level of access (user/admin) an attacker could gain if exploited.

4. Why are Windows Null Sessions considered a security weakness, and how would you assess their impact?

Answer:

Null sessions allow anonymous, unauthenticated connections to a system, letting attackers enumerate usernames, shares, and password policies without any credentials. Impact is assessed by how much sensitive information (user lists, share names, security policy) is exposed through this channel.

5. How would you distinguish between a vulnerable Windows service and a misconfigured Windows service during security testing?

Answer:

A vulnerable service has an actual flaw in its code or an outdated version that needs patching, while a misconfigured service is otherwise secure software that has been set up incorrectly (e.g., weak permissions, default credentials) and needs reconfiguration, not patching.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark
Understanding of Problem Context	20	Clearly understands the security problem and accurately identifies vulnerabilities, risks, and requirements	Good understanding of the security problem with minor gaps	Basic understanding; some security requirements or vulnerabilities are unclear	Poor understanding or misinterpretation of the security problem	
Application of Concepts	30	Accurately applies appropriate ethical hacking concepts, tools, and exploitation techniques to analyze complex security problems	Applies relevant security concepts and tools correctly with minor errors	Applies basic concepts with limited accuracy or tool selection	Fails to apply appropriate security concepts or tools	
Problem Solving Approach	20	Demonstrates a logical, well structured, systematic, and efficient approach to identifying and evaluating security weaknesses	Mostly logical and systematic approach with minor gaps	Basic approach with some inconsistencies in vulnerability analysis	Illogical, unclear, or inappropriate approach to solving the security problem	

Accuracy of Solution	20	Provides a completely correct security analysis with appropriate justification, evidence, and suitable prevention	Mostly correct analysis with minor errors or gaps	Partially correct analysis with limited justification	Incorrect or incomplete security analysis	
Clarity	10	techniques Provides clear, well-organized, technically sound, and properly explained responses using appropriate security terminology	Understandable response with minor clarity or terminology issues	Limited clarity and explanation; uses technical terms inconsistently	Poorly presented, unclear, and difficult to understand	
Total	100	terminology				/100